

The Hubble-Kernel Growth Equation Against DESI DR1: Consistent, Not Yet Discriminated, and the Precision That Decides

Carl P. Zimmerman

Briar Creek Tech

AI-assisted research programme; not peer reviewed

11 September 2026

Abstract

The Hubble-kernel growth equation (doi:10.5281/zenodo.22706925) fixes the effective gravitational coupling of linear structure with no free parameter, $G_{\text{eff}}(z)/G = \nu(cH(z)/a_0)$ with $(cH_0/a_0)^2 = 8\pi/(3\kappa^2\Omega_\Lambda) = 49$, and predicts $f\sigma_8$ raised over Planck- Λ CDM by 2.7%, 1.9%, 1.1% at $z = 0.3, 0.6, 1.0$ (canonical footing; 3.8, 2.6, 1.6% on the alternative footing). We confront it with the DESI DR1 full-shape growth measurements (DESI 2024 V, Table 9: ShapeFit $f\sigma_{ss}/(f\sigma_{ss})_{\text{fid}}$ in six redshift bins, Planck fiducial, both the ShapeFit-only and ShapeFit+BAO sub-panels) and with the DESI modified-gravity parameter $\mu_0 = 0.05 \pm 0.22$. The equation is consistent with the data (χ^2 per bin 0.65–0.73 on both footings and both sub-panels) and is not yet discriminated from Λ CDM ($|\Delta\chi^2| < 0.25$): DESI DR1’s per-bin uncertainties of 10–19% cannot resolve a 1–4% effect. The equation’s present-day coupling excess (0.077 canonical, 0.099 alternative) lies 0.1–0.2 σ from μ_0 . A 3 σ test requires $\sim 1\%$ on $f\sigma_8$ at $z \simeq 0.3$ –0.6, the DESI final-sample regime. This is a status result: the prediction is alive, on record before the decisive data, and its sign (more growth, not less) is the one the S_8 tension does not favour.

1 The prediction

With $a_0 = \kappa c\sqrt{G\rho_\Lambda}$ and the Hubble-flow acceleration as the kernel’s cosmological ambient field, $G_{\text{eff}}(z)/G = \nu(cH(z)/a_0)$, $\nu(x) = 1/(1 - e^{-\sqrt{x}})$, $cH_0/a_0 = 7.0$ ($\kappa = \frac{1}{2}$; 5.8 on the alternative footing $a_0 = 1.13 \times 10^{-10} \text{ m s}^{-2}$). Linear growth with this coupling on a Λ CDM background ($\Omega_m = 0.314$, $h = 0.674$) gives the ratio of $f\sigma_8$ to its Planck- Λ CDM value listed in Table 1.

2 The data

DESI 2024 V [1] reports, for each of six redshift bins, the ShapeFit compressed parameter $f\sigma_{ss}/(f\sigma_{ss})_{\text{fid}}$ (fiducial: Planck 2018 Λ CDM; MAP values with 68% credible intervals), in two sub-panels: ShapeFit only, and ShapeFit with the post-reconstruction BAO information added. We symmetrise asymmetric intervals as the mean of the two sides. Separately, DESI 2024 [2] constrains the phenomenological coupling $\mu(a) = 1 + \mu_0 \Omega_\Lambda(a)/\Omega_\Lambda$ to $\mu_0 = 0.05 \pm 0.22$ from DESI full-shape and BAO combined with CMB, DES Y3 lensing and DES Y5 supernovae.

3 Confrontation

Table 1: DESI DR1 ShapeFit growth ratios versus the equation.

bin	z_{eff}	ShapeFit only	ShapeFit+BAO	eq. (canonical)	eq. (alt)
BGS	0.295	0.80 ± 0.20	0.84 ± 0.19	1.027	1.038
LRG1	0.510	$1.09^{+0.12}_{-0.14}$	1.16 ± 0.13	1.021	1.029
LRG2	0.706	1.05 ± 0.12	$1.04^{+0.11}_{-0.09}$	1.016	1.023
LRG3	0.919	$0.96^{+0.11}_{-0.10}$	$1.00^{+0.10}_{-0.08}$	1.012	1.018
ELG2	1.317	$0.95^{+0.11}_{-0.08}$	$0.95^{+0.10}_{-0.08}$	1.007	1.011
QSO	1.491	1.16 ± 0.12	1.16 ± 0.12	1.006	1.009

Over the six bins: $\chi^2 = 3.85$ (Λ CDM), 3.92 (canonical), 3.98 (alt) for ShapeFit only; 4.56, 4.36, 4.33 for ShapeFit+BAO. The equation is consistent with the data ($\chi^2/6 = 0.65\text{--}0.73$) and undiscriminated from Λ CDM ($\Delta\chi^2$ between -0.23 and $+0.13$). Its present-day coupling excess, $G_{\text{eff}}/G - 1 = 0.077$ (canonical) or 0.099 (alt), lies 0.12σ and 0.22σ from μ_0 .

4 What decides

The predicted effect is 1–4% at $z < 1$; DESI DR1 resolves 10–19% per bin. A 3σ test needs $\sim 1\%$ on $f\sigma_8$ at $z \simeq 0.3\text{--}0.6$, which is the reach of the DESI final sample. A measured deficit of growth at the few-percent level, or a weak-lensing S_8 two to three percent below Planck- Λ CDM, falsifies the equation; an excess of the predicted size confirms it against Λ CDM. The prediction is on record before either. Provenance: repository lanes L180–L181 (`fable_independent_2026/`, committed scripts and outputs); the equation’s derivation and scope are in doi:10.5281/zenodo.22706925.

References

- [1] DESI Collaboration, *DESI 2024 V: Full-Shape Galaxy Clustering from Galaxies and Quasars*, arXiv:2411.12021.
- [2] M. Ishak et al. (DESI Collaboration), *Modified Gravity Constraints from the Full Shape Modeling of Clustering Measurements from DESI 2024*, arXiv:2411.12026.